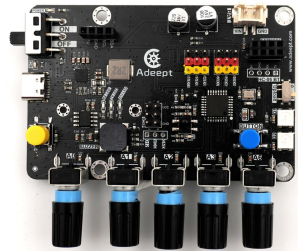


Lesson 5 OLED Display

5.1 Overview

This course mainly introduces the application of OLED display screen on Adeept Robotic Arm Control Board, covering the required components, working principle, hardware connection, code writing and demonstration, to help master the combination of OLED display screen and Arduino, and achieve the display function of information on OLED screen.

5.2 Required Components

Components	Quantity	Picture
Adeept Robotic Arm Control Board	1	
Type-C USB Cable	1	
OLED Screen	1	

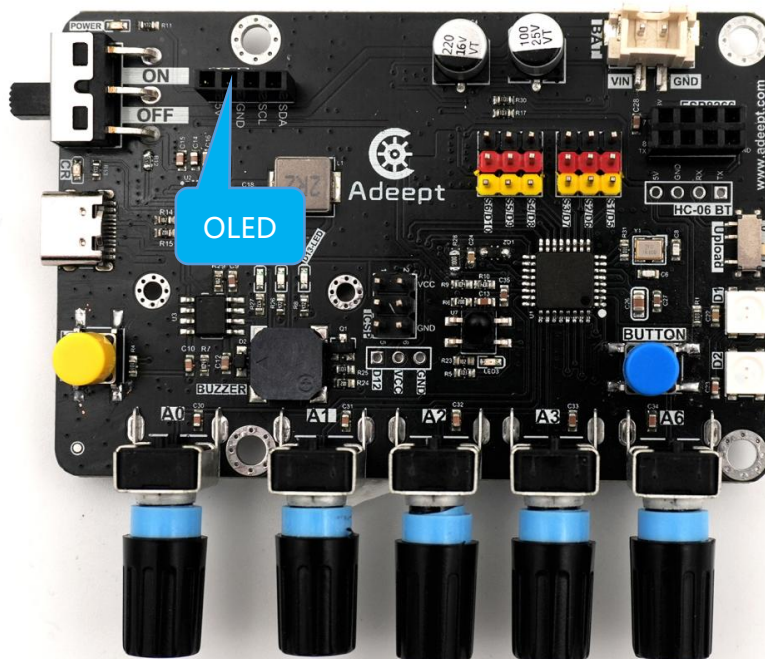
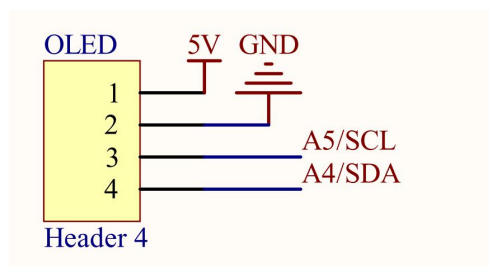
5.3 Principle Introduction

OLED (Organic Light-Emitting Diode), also known as organic electroluminescence display or organic light-emitting semiconductor, belongs to a current-type organic light-emitting device. It

emits light through the injection and recombination of carriers, and the luminous intensity is proportional to the injected current, with relatively low power consumption.

Due to their small size, light weight, and low power consumption, OLED screens are increasingly used in embedded electronic devices. There are various types of OLED screens, which display different colors. Commonly used ones include white display, blue display, and yellow-blue two-color display. Additionally, they come in different screen sizes and are equipped with various built-in driver chips. The commonly used driver interfaces are SPI and IIC. This article introduces a 0.96-inch blue IIC-driven screen, and its built-in driver chip is SSD1306.

5.4 Wiring Diagram



PINS of Adeept Robotic Arm Control Board	OLED
A4	SDA
A5	SCL
GND	GND
5V	VCC

5.5 Demonstration

1. Connect your computer and Adeept Robotic Arm Control Board with a USB cable.
2. Open "**05_OLED**" folder in **"/Code"** , double-click "**05_OLED.ino**" .

```

05_OLED | Arduino IDE 2.3.10
File Edit Sketch Tools Help
[Icons] Select Board [Icons]

05_OLED.ino
1  #include <Wire.h>
2  #include <SSD1306AsciiWire.h>
3
4  #define SCREEN_W  128
5  #define SCREEN_H  64
6  #define OLED_ADDR 0x3C
7
8  SSD1306AsciiWire display;
9
10 void setup() {
11     Serial.begin(115200);
12
13     Wire.begin();
14     display.begin(&Adafruit128x64, OLED_ADDR);
15     display.setFont(Adafruit5x7);
16     display.clear();
17
18     display.set2X();
19     display.setCursor(20, 2);
20     display.print("Adeept");
21 }
22
23 void loop() {
24
25 }

```

3. Select development board and serial port.

Board: **Tools**--->**Board**--->**Arduino AVR Boards**--->**Arduino Uno**

Port: **Tools** --->**Port**--->**COMx**

Note: The port number will be different in different computers.



4. After opening, click  to upload the code program to the Arduino. If there is no error warning in the console below, it means that the Upload is successful.

```
Output
Sketch uses 2952 bytes (9%) of program storage space. Maximum is 32256 bytes.
Global variables use 82 bytes (4%) of dynamic memory, leaving 1966 bytes for local variables. Maximum is 2048 bytes.
```

5. When the program runs successfully, you will clearly see the OLED displays "Adeept" .

5.6 Code

```
01 #include <Wire.h>
02 #include <SSD1306AsciiWire.h>
03
04 #define SCREEN_W 128
05 #define SCREEN_H 64
06 #define OLED_ADDR 0x3C
07
08 SSD1306AsciiWire display;
09
10 void setup() {
11     Serial.begin(115200);
12
13     Wire.begin();
14     display.begin(&Adafruit128x64, OLED_ADDR);
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20     display.print("Adeept");
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23 void loop() {
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25 }
```